

Alias & functions

Typing the same long command again and again is slow

Shortcut vs helper with args

- An alias expands to text before the shell runs the line
- Aliases are awkward with real arguments; prefer a function when you need ``$1`` or ``"$@"``
- Functions run in the current shell, so they can change directory or set variables for you
- Scripts run in a subshell, better when you want a file to share and version
- Session aliases and functions vanish when the shell exits unless you add them to your

Browser lab

Digital Design and Verification Platform

HomeTools

HomeToolsAlias / function

Contrast shell alias shortcuts with functions that take arguments – and see what is session-only vs what you would put in .bashrc.

Starter example: alias ll='ls -la' and greet() { echo hello \$1; } are defined. Run ll vs greet lab – aliases don't take real args the same way.

Load starter example

Challenges 0 / 22 cleared

Quiz: alias: An alias is mainly a? Answer: shortcut or substitution

Answer

Show hintCheckNextIdle

Quiz: aliasRun llList aliasesQuiz: \$!greet labtype lltype greetDefine aliasumalias

Quiz: sessionsave-rcnew-shell keepsnew-shell losesDefine functionQuiz: prefer fnQuiz: bashrc

alias aliasQuiz: expandShow oneaddpathQuiz: bothbashrc content

Lab shell

Session only: aliases/functions vanish on new-shell unless you save-rc first.

Starter: alias ll / sim . function greet - try: ll . greet lab - type ll - save-rc - new-shell

ll\$ alias - unalias - name() { - ? - type - help

alias llgreet labtype lltype greetalias g~

umalias simsave-rcnew-shell

Defined names

kind	name	body
alias	ll	ls -la
alias	sim	echo run_sim
function	addpath	echo PATH=\$1
function	greet	echo hello \$1

Would go in ~/.bashrc

not saved yet - run save-rc

Alias vs function

alias

- Text substitution of the command word
- Awkward with arguments / logic
- Session-only unless saved in .bashrc

function

- Real body: \$1, \$2, ...
- Better for multi-step / conditionals
- Also session-only until sourced from a file

Cheat sheet

Command	Meaning
alias name="cmd"	Define alias
alias	List aliases
unalias name	Remove alias
name() { ... }	Define function (lab one-liner)
type name	Show alias / function / command
save-rc	Lab: pretend write defs to ~/.bashrc
new-shell	Lab: new session (keeps only "saved" defs)

Typing alias ll after defining it shows the expansion.

Prefer functions when you need arguments or more than a short shortcut.

Alias lab

learn_unix — alias lab

Real shell practice

```
wsl - module15-alias-lab/examples/aliases

# Track A - real Unix
# real shell session (Track A)

$ alias ll='ls -alF'

$ ll
total 4
drwxrwxrwx 1 yongfu yongfu 4096 Jul 18 16:34 ./
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 00:49 ../
-rwxrwxrwx 1 yongfu yongfu 1290 Jul 18 16:34 README.md*
-rwxrwxrwx 1 yongfu yongfu  368 Jul 18 16:34 commands_to_try.txt*

$ alias | grep ll
alias ll='ls -alF'

$ greet() { echo Hello "$1"; }

$ greet Alice
Hello Alice

$ type greet
greet is a function
greet ()
{
    echo Hello "$1"
}

$ unalias ll
```

Real shell — alias and function

Real shell practice — try these

```
# alias ll='...' — define a session alias for a long listing
alias ll='ls -alF'
```

```
# ll — run the alias (expands to ls -alF)
ll
```

```
# alias | grep ll — list aliases and keep lines matching ll
alias | grep ll
```

```
# greet() { ... } — define a function that uses the first argument
greet() { echo Hello "$1"; }
```

```
# greet Alice — call the function with one argument
greet Alice
```

```
# type greet — show whether greet is an alias, function, or command
```

Pitfalls to watch

- Do not put spaces around the equals in ``alias name='cmd'``
- Remember aliases do not take arguments the way functions do
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, define an alias and a small function, then try ``type`` and ``unalias``
- When you are ready, take the short quiz, then continue to script control flow